

Day 12 of training

Javascript

What is a Loop?

A loop is used to repeat a block of code again and again until a condition is false.

1). For loop :- A for loop repeats a block of code a fixed number of times.

Syntax:-

```
for (initialization; condition; update) {  
    // code to repeat  
}
```

2). while loop

A while loop keeps running as long as the condition is true.

Syntax:-

```
let i=1;  
  
while(i<5){  
    document.write(i+'<br>');  
    i++;  
}
```

3). What is a do...while loop?

A do...while loop executes the block of code at least once, and then checks the condition.

```
let i=10;  
  
do{  
    document.write(i+'<br>');  
    i++;  
}while(i<=5);
```

What is break?

The break statement is used to stop a loop immediately, even if the condition is still true.

What is continue?

The continue statement skips the current iteration of the loop and jumps to the next one, without stopping the whole loop.

What is an Array in JavaScript?

An array is a special variable in JavaScript that can store multiple values in a single variable.

Key Points:

Arrays can store any type: numbers, strings, booleans, objects, even other arrays!

They are written using square brackets [].

Values are separated by commas ,.

What is a Method in JavaScript?

A method is just a function that is attached to an object.

In Simple Words:

Method = Function inside an object

```
let person = {  
  name: "Opinder",  
  greet: function () {  
    console.log("Hello, I'm " + this.name);  
  }  
};
```

```
person.greet(); // Output: Hello, I'm Opinder
```

Here: `greet` is a method of the `person` object.

It behaves like a normal function but is called using the object.

Array Method

`. push()` – Add at the end

Adds something to the end of the array.

`pop()` – Remove from end

Removes the last item.

`unshift()` – Add at start

Adds item at the beginning.

`shift()` – Remove from start

Removes the first item.

`includes()` – Check if item exists

Checks if an item is there or not.

`indexOf()` – Find position

Tells where the item is.

`join()` – Convert array to string

Joins everything with comma (or your own symbol).

`slice()` – Copy part of array

Cuts a piece of array (original not changed).

`splice()` – Cut/Insert items

Can remove or add items (original array changes).

-->Object in JavaScript:

An object is a collection of key-value pairs where:

Keys are called properties

Values can be data or functions (methods)

```
let car = {
```

```

brand: "Toyota", // property
color: "red",    // property
start: function() { // method
    console.log("Car started");
}
};

```

MATH METHODS (Built-in Math object)

Math methods are used to do calculations.

Method	Meaning	Example	Result
Math.round()	Rounds to nearest	Math.round(4.6)	5
Math.floor()	Rounds down	Math.floor(4.9)	4
Math.ceil()	Rounds up	Math.ceil(4.1)	5
Math.max()	Largest number	Math.max(2, 7, 5)	7
Math.min()	Smallest number	Math.min(2, 7, 5)	2
Math.random()	Random 0 to 1	Math.random()	0.123...
Math.pow()	Power (x^n)	Math.pow(2, 3)	8
Math.sqrt()	Square root	Math.sqrt(9)	3
Math.abs()	Absolute (no -ve)	Math.abs(-7)	7

DATE METHODS (Using new Date())

Used to work with dates and times.

Method	Meaning	Example	Output (Example)
-----	-----	-----	-----
<code>`getFullYear()`</code>	Get year	<code>`now.getFullYear()`</code>	2025
<code>`getMonth()`</code>	Get month (0–11)	<code>`now.getMonth()`</code>	6 (July)
<code>`getDate()`</code>	Get day of month	<code>`now.getDate()`</code>	14
<code>`getDay()`</code>	Day of week (0–6)	<code>`now.getDay()`</code>	1 (Mon)
<code>`getHours()`</code>	Get hours	<code>`now.getHours()`</code>	20
<code>`getMinutes()`</code>	Get minutes	<code>`now.getMinutes()`</code>	32
<code>`getSeconds()`</code>	Get seconds	<code>`now.getSeconds()`</code>	10
<code>`toDateStrinɡ()`</code>	Date in string	<code>`now.toDateStrinɡ()`</code>	"Mon Jul 14 2025"
<code>`toTimeStrinɡ()`</code>	Time in string	<code>`now.toTimeStrinɡ()`</code>	"20:32:10 GMT+0530..."